18 OCT 2023

Reg. No.:_______________

Amrita Vishwa Vidyapeetham
Amrita School of Computing, Bengaluru
B.Tech. Mid-term exam – Oct. 2023
Computer Science and Engineering
Seventh Semester

19CSE353 - Mining of Massive Datasets

Time: Two hours | Maximum marks: 50

CO	Course Outcomes
CO1	Understand the importance of data to business decisions, strategy and behavior. Predictive analytics, data mining and machine learning as tools give new methods for analyzing massive data sets.
CO2	Expose the students to Big data systems like Hadoop, Spark and Hive.
CO3	Explore means to deal with huge document databases and infinite streams of data to mining large social networks and web graphs. Also, learn Algorithms suitable for large scale mining.
CO4	Use of analytic tools, case-studies to provide first hand insight into how big-data problems and their solutions are commercially derived.
CO5	Design Large Scale Machine Learning algorithms with Practical hands-on experience for analyzing very large amounts of data

Answer all questions

1. Suppose we implement the Map Reduce program on a large repository of documents belonging to 'Resume', 'Cricket' and 'University' categories. The aim of the Map reduce task is to compute the frequency of different alphabets (small letters, capital letters and numeric digits). Hence, we plan to use 100 Map tasks and few number of Reduce tasks Answer the following questions based on the given scenario:

a. Explain the Map, Combine and Reduce functions with map-reduce process diagram. [4] CO2 BTL-2

b. We do not use a combiner at the Map tasks and run only one reducer per Reduce task node. Each reducer is responsible for an individual character. Examine will there be significant skew in the time taken by the various Reduce tasks to process their value list? Justify. [3] CO2 BTL-4

c. If we combine multiple reducers into a small number of Reduce tasks, say 5 tasks, at random and one reducer is responsible for multiple characters. Examine will there be significant skew in the time taken by the various Reduce tasks to process their value list? Justify. [3] CO2 BTL-4

2. a. Explain power law with expression? Given a company 'A' releases around 100 products and each product is rated according to its popularity/ sale. Explain relation between the sale/ profit earned and the product popularity using Power law with diagram. [2+3] CO1 [BTL 2]

3 Consider the following 4 documents D1, D2, D3 and D4.

D1 : I am Ram.
D2 : Ram I am.
D3 : I do not like softy and chips.
D4 : I do not like them, Ram I am.
We plan to create k-shingles at word level with k=1 for all documents. For instance Shingles at word-level for D1 with k=1 are {I, am, Ram}. In this way generate the set 'S' of all unique word-level 1-shingles for all the documents. In 'S' arrange shingles in alphabetical order.

a. Create the two dimensional Boolean matrix representation depicting relations between each shingle and individual document. Arrange shingles in alphabetical order. [3.5] CO3 [BTL-2]

b. Generate min-hash signatures (matrix) of length 3 for each document using given row permutations on Boolean matrix. [1.5*3] CO3 [BTL-3]

P1: 2, 4, 6, 8,10, 1, 3, 5, 7, 9, 11

P2: 1, 3, 5, 7, 11, 2, 4, 6, 8, 9, 10

P3: 1, 4, 7, 10, 2, 5, 8, 11, 3, 6, 9

c. With the help of 'Jaccard similarity' compute the similarity between the documents {D1, D3} and {D2, D4} using min-hash signatures and Boolean matrix separately and compare them. [2+2] CO1 [BTL-4]

4. a. What is a standing query and an adhoc query in stream processing? [1.5 *2] CO1 BTL-1

a. Given that you are a Youtube content creator and uploader, you would like run few queries to check the popularity of the each content uploaded by you. Suppose following are queries that you would like to ask. Which of these queries are standing and ad-hoc queries? [4] CO3 [BTL-3]

i. Notify when a particular video 'V1' reaches 1 million views.

ii. Notify what is the sentiment of the users based on the comments posted for video 'V1' over past two months

iii. Notify the sentiment of the users based on the number of 'Likes' and 'Dislikes' received for a video 'V1'

iv. Notify the videos watched by different users repeatedly over past one month.

5 Answer the following questions with respect to the stream sampling

a.	State the property to be maintained by the samples of a stream in time.	[2]	CO3 BTL-1
b	What is the issue in using fixed-proportion sampling technique with constrained memory? How do you handle it?	[2]	CO3 BTL-1

6	a.	How do you compute the relevance of web pages? Discuss with example.	[1+1]	CO3 BTL-1
	b.	.The web can be represented as a directed graph where nodes represent the web pages and edges form links between them. Calculate the importance score (page rank) of each node in the graph given in Fig.	[2 * 5]	CO3 [BTL-3]

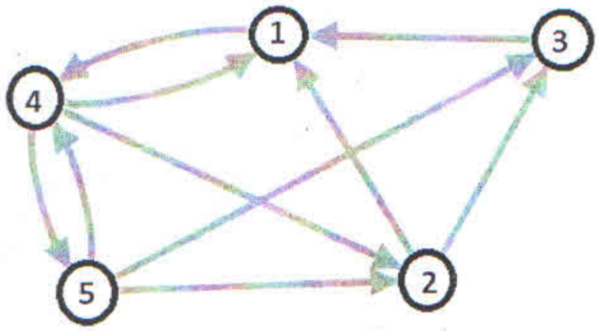

Fig 1: Graph for Q 6b

Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	12	BTL-1	9
CO2	10	BTL-2	12.5
CO3	28	BTL-3	18.5
CO4	-	BTL-4	10
CO5	-	BTL-5	-